

2024



instituto de ingeniería del conocimiento

IIC Powers AI Predictions by Going Serverless on AWS



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Learn how Instituto de Ingenieria del Conocimiento delivers AI-powered predictions at scale using AWS serverless technologies.



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30% increase

in AI prediction accuracy

90% decrease

in monitoring and tracing effort

20 Million

new predictions made per year

Overview

For nearly 20 years, [Instituto de Ingeniería del Conocimiento \(IIC\)](#) has provided renewable energy prediction services to its clients. The research and development center specializes in artificial intelligence (AI) and other advanced technologies, helping energy companies better understand and optimize their operations. However, its engineers faced difficulties providing this service due to an outdated on-premises system.

To deliver AI predictions at scale, IIC completely redesigned its architecture by migrating to Amazon Web Services (AWS) and adopting an event-driven architecture (EDA) powered by serverless technologies. By making this change, the research and development center could develop a new application: the EA3 renewable energy prediction system. EA3 has provided value to multiple renewable energy companies, delivering accurate predictions of energy production in wind farms and photovoltaic, solar thermal, and hydroelectric plants.



Opportunity | Using AWS Serverless Technologies to Modernize a Monolithic On-Premises System for IIC

Based in Spain, IIC has over 30 years of experience in big data analysis, machine learning, and natural language processing. In 2021, the company confronted a critical challenge that affected its ability to deliver AI predictions at scale. “There was an outdated and monolithic on-premises system that made its operation, improvement, and scalability difficult,” says Álvaro Romero, product owner of EA3 at IIC. “There was a pressing need for modernization so that we could expand the application in an increasingly competitive market, where prediction accuracy is a key differentiator.”

To confront these challenges, IIC embarked on a major system redesign. It chose to migrate its system to AWS and implement an EDA, which relies on serverless functions to dynamically respond to near real-time events by



activating specific, automated actions. “The new system not only facilitated more efficient and better-quality operations but also provided the ability to incorporate multiple new data sources and adapt to new use cases,” says Romero.

IIC chose AWS for EA3’s deployment due to its comprehensive service range and maturity. The company could meet increased data processing demands and scale up resources by adopting a serverless operation model on AWS, particularly the automatic scaling feature of [AWS Lambda](#)—a serverless, event-driven compute service. As a result, the company could facilitate optimal performance and continued customer satisfaction. “Our highly redundant and fault-tolerant architecture on AWS provides reliable delivery of events and messages,” says Romero. “This reduces the risk of interruptions and facilitates continuous service for users.”

Solution | Increasing AI Prediction Accuracy by 30% with EDA

Thanks to the redesign, IIC can now efficiently process diverse data sources through EA3. For effective communication within the architecture, IIC relies on [Amazon Simple Notification Service](#) (Amazon SNS), a fully managed Pub/Sub service for A2A and A2P messaging, and [Amazon Simple Queue Service](#) (Amazon SQS), a fully managed message queuing for microservices, distributed systems, and serverless applications. The company uses Amazon SNS to manage notifications and Amazon SQS to facilitate asynchronous data delivery, while AWS Lambda is used to run scalable data processing and AI inference.

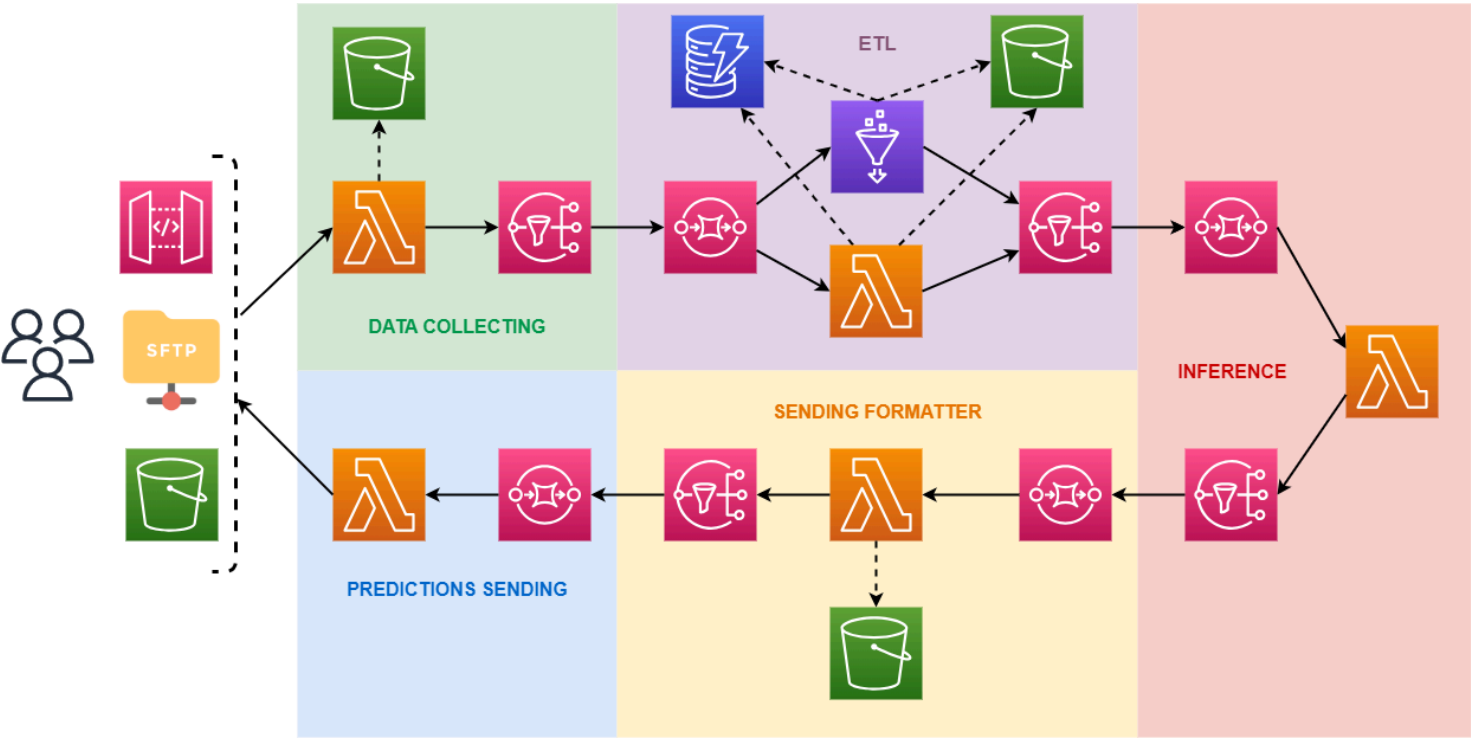
To transform data, EA3 relies on serverless functions in AWS Lambda to carry out specific tasks and scale the architecture as needed. For complex and heavy transformations, the system uses [AWS Glue](#), a service that is used to discover, prepare, and integrate all data at any scale. IIC relies on AWS storage and database services, such as [Amazon Simple Storage Service](#) (Amazon S3)—an object storage service built to retrieve any amount of data from anywhere—to uphold data integrity and facilitate efficient access to it.

While developing EA3, IIC placed major emphasis on flexibility and the ability to integrate data sources and external systems. The use of AWS services has been essential to achieve this versatility, helping IIC adapt EA3 to meet the needs of each energy client. The system facilitates seamless data exchange using services such as [AWS Step Functions](#), a visual workflow service, and [Amazon API Gateway](#), which is used to create, manage, and secure APIs at any scale.

With an EDA powered by AWS, IIC can integrate even more versatile data sources, which include weather forecasts and power plant variables for precise renewable energy predictions. The company can also scale data transformation and processing as needed. Due to enhanced data sources, the company has increased EA3's AI prediction accuracy by 30 percent. By adopting a wide range of observability tools from AWS, IIC has also reduced the operation time required for tracing and monitoring by 90 percent.

Architecture Diagram

IIC's EDA on AWS



Outcome | Continuing to Evolve EA3 with Technologies from AWS

In the future, IIC plans to continually enhance EA3 for optimal agility and performance. It plans to add new features powered by AWS technologies and explore new use cases, such as incorporating new data sources and tailoring the tool for its clients' specific needs. As it moves forward, IIC is committed to maintaining its position in the renewable energy sector, evolving EA3 to meet dynamic market demands. And with serverless technologies from AWS, the company is poised to swiftly adapt and scale.

"The new system currently makes more than 20 million new predictions per year," says Romero. "Thanks to the great scalability that we enjoy on AWS, this figure can continue to grow as more clients join."

About Instituto de Ingenieria del Conocimiento

Instituto de Ingenieria del Conocimiento (IIC) is a research and development center specializing in artificial intelligence. Based in Spain, IIC holds more than 30 years of experience in big data analysis, machine learning, and natural language processing.

AWS Services Used

AWS Lambda

AWS Lambda is a serverless, event-driven compute service that lets you run code for virtually any type of application or backend service without provisioning or managing servers.

[Learn more »](#)

Amazon SNS

Amazon Simple Notification Service (Amazon SNS) is a fully managed messaging service for both application-to-application (A2A) and application-to-person (A2P) communication.

[Learn more »](#)

Amazon SQS

Amazon Simple Queue Service (Amazon SQS) lets you send, store, and receive messages between software components at any volume, without losing messages or requiring other services to be available.

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AWS Glue

AWS Glue is a serverless data integration service that makes data preparation simpler, faster, and cheaper.

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